PS3: ARCH, GARCH

Michael Fehl, Francesco Zucca, Giuseppe Lista

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```r
library(tidyverse)   # tidy data ecosystem
library(patchwork)   # patch ggplots
library(fGarch)      # ARCH, GARCH modeling
library(fBasics)     # basicStats()
```

1 Introduction

In this practice, we will work on fitting different volatility models (ARCH and GARCH). We will also focus on discussing the validation and selection of the model. As done in previous practices, each student group must follow the practice script below, performing the necessary calculations and graphs in the R environment. The obtained results and answers to the questions should be compiled into a document containing your solution, created, for example, in R Markdown or Microsoft Word. Please make sure to include your full names and group number at the beginning of the document. Submit the questionnaire solution in PDF format by uploading it to the platform under the corresponding task section before the deadline.

In this questionnaire, we will work with the following time series:

- KO: The Coca-Cola Company (KO) stock (NYSE)

You can download the data using this code:

```r
library(quantmod)   # quant finc modeling and trading framework
getSymbols("KO")
```

```
[1] "KO"
```

2 Questions

2.1 Download Data

Task: Download the time series. You may work with the complete dataset (from 2007 to the most recent trading session).

Solution

```r
cat('column names:', names(KO), '\n\n')  # columns
```

```
column names: KO.Open KO.High KO.Low KO.Close KO.Volume KO.Adjusted
```

```r
summary(KO)                              # quick summary stats
```

```
     Index               KO.Open         KO.High          KO.Low     
 Min.   :2007-01-03   Min.   :19.08   Min.   :19.61   Min.   :18.72  
 1st Qu.:2011-10-13   1st Qu.:33.97   1st Qu.:34.21   1st Qu.:33.77  
 Median :2016-08-01   Median :42.82   Median :43.20   Median :42.53  
 Mean   :2016-08-01   Mean   :44.55   Mean   :44.86   Mean   :44.23  
 3rd Qu.:2021-05-17   3rd Qu.:54.60   3rd Qu.:54.91   3rd Qu.:54.26  
 Max.   :2026-03-06   Max.   :81.25   Max.   :82.00   Max.   :80.82  
    KO.Close        KO.Volume           KO.Adjusted   
 Min.   :18.92   Min.   :  2996300   Min.   :11.18  
 1st Qu.:34.00   1st Qu.: 11417600   1st Qu.:21.87  
 Median :42.92   Median : 14403150   Median :31.81  
 Mean   :44.55   Mean   : 16210680   Mean   :35.44  
 3rd Qu.:54.61   3rd Qu.: 18930075   3rd Qu.:47.46  
 Max.   :81.56   Max.   :124169000   Max.   :81.56  
```

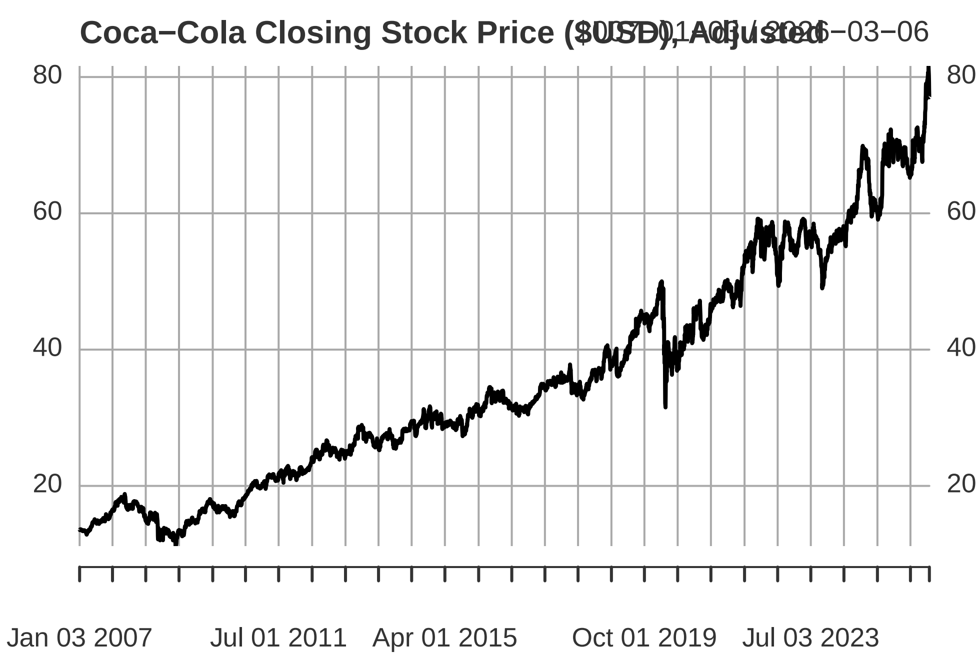
```r
chartSeries(KO)                          # financial chart
```

2.2 Graphical Representation

Task: Create a graphical representation of the time series using the Adjusted Closing Price.

Solution

```
plot(KO$KO.Adjusted,
     main = "Coca-Cola Closing Stock Price ($USD), Adjusted")
```



2.3 Compute Log-Returns

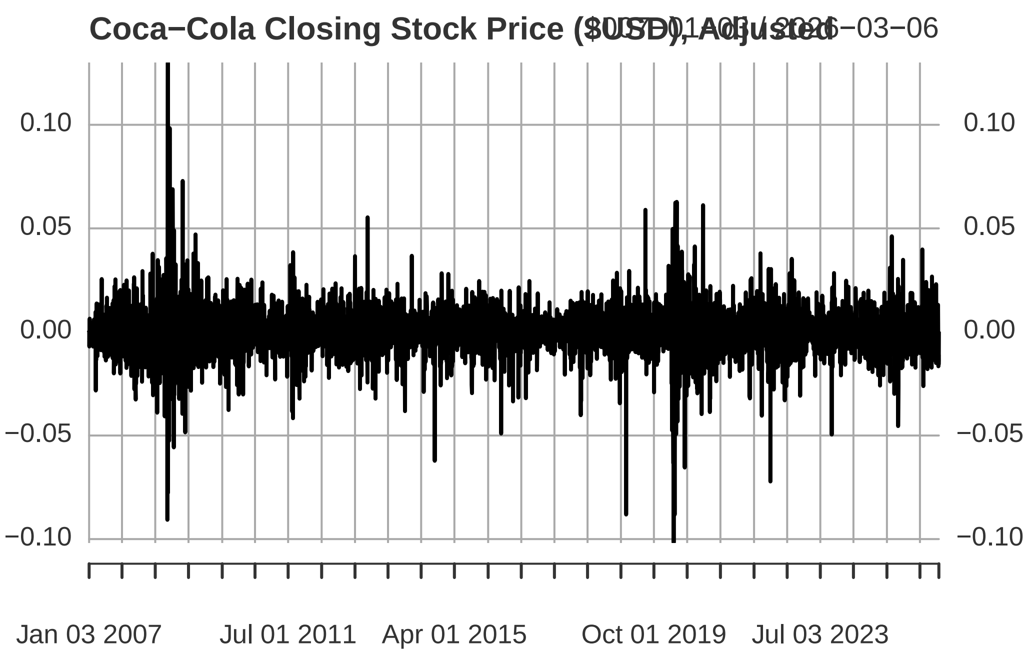
Task: Compute the log-returns of the series. It is not necessary to model the mean (e.g., using an ARIMA model).

Solution

```
# returns: can do this directly using fQuant library
KO$log_returns <- dailyReturn(KO$KO.Adjusted,
                              type = "log")

# sanity check
head(KO$log_returns)
```

```
             log_returns
2007-01-03  0.0000000000
2007-01-04  0.0004118395
2007-01-05 -0.0070204185
2007-01-08  0.0064024905
2007-01-09  0.0008230482
2007-01-10  0.0014397037
```

```
# plot
plot(KO$log_returns,
     main = "Coca-Cola Closing Stock Price ($USD), Adjusted") # looks stationary -- good
```



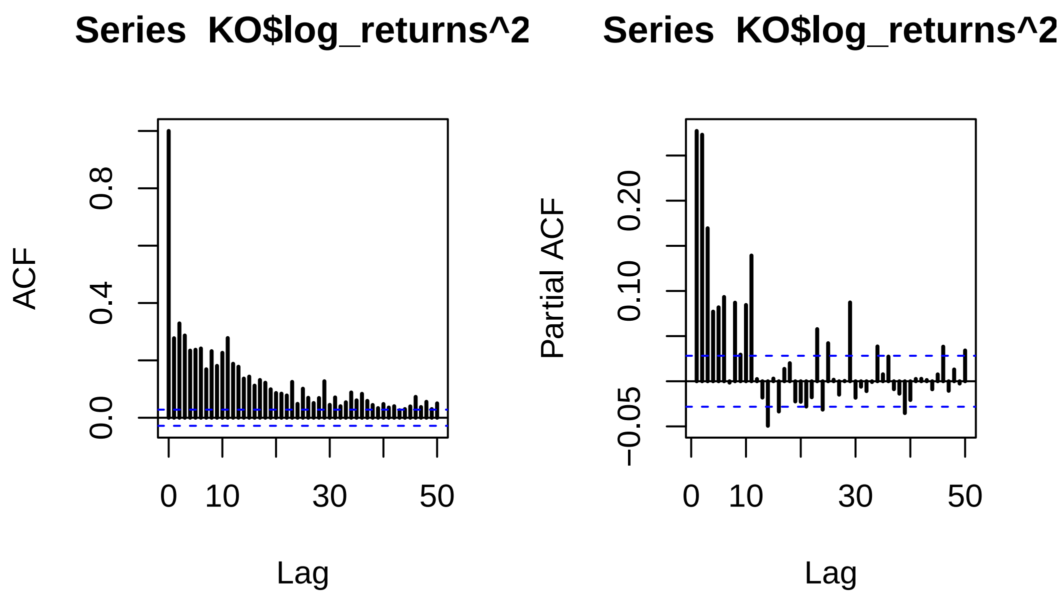
Before moving on to model fitting, we do some basic analysis of the returns.

```
stats <- basicStats(KO$log_returns)
stats[c("Skewness", "Kurtosis"), , drop = FALSE]
```

```
         log_returns
Skewness   -0.160944
Kurtosis   11.303078
```

The time series is more skewed compared to the normal distribution, and it has fatter tails because of the excess kurtosis of around 11.

```r
par(mfrow = c(1, 2))
acf(KO$log_returns^2, lwd = 2, lag.max = 50)
pacf(KO$log_returns^2, lwd = 2, lag.max = 50)
```




```r
par(mfrow=c(1,1))
```

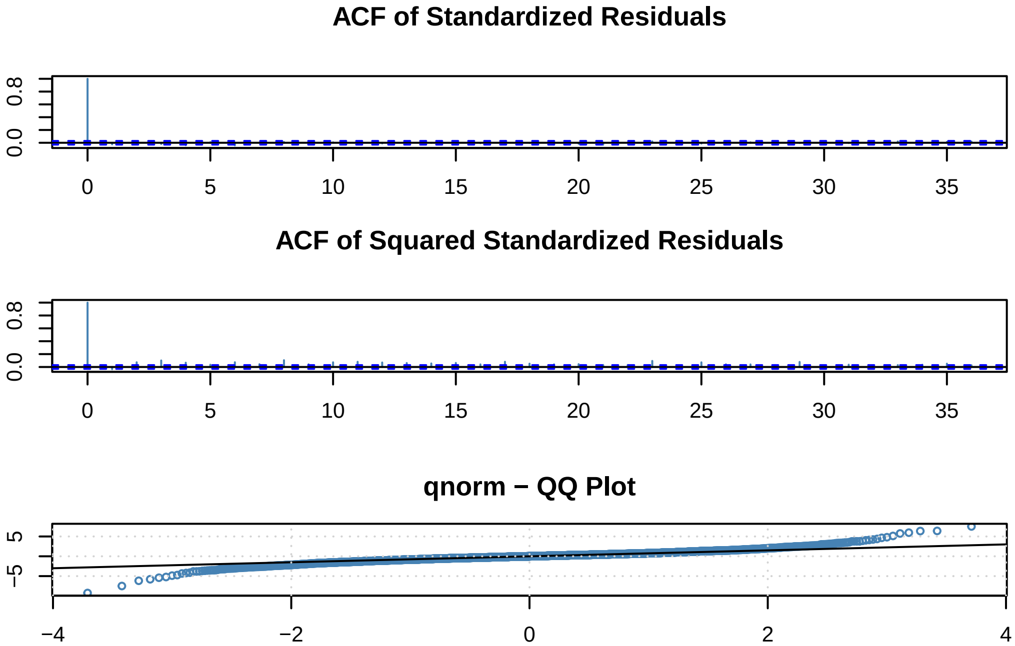
2.4 Model Fitting

Task: Using the fGarch package, fit the following models to the log-returns: an ARCH(1) model, an ARCH(3) model, and a GARCH(1,1) model.

Solution

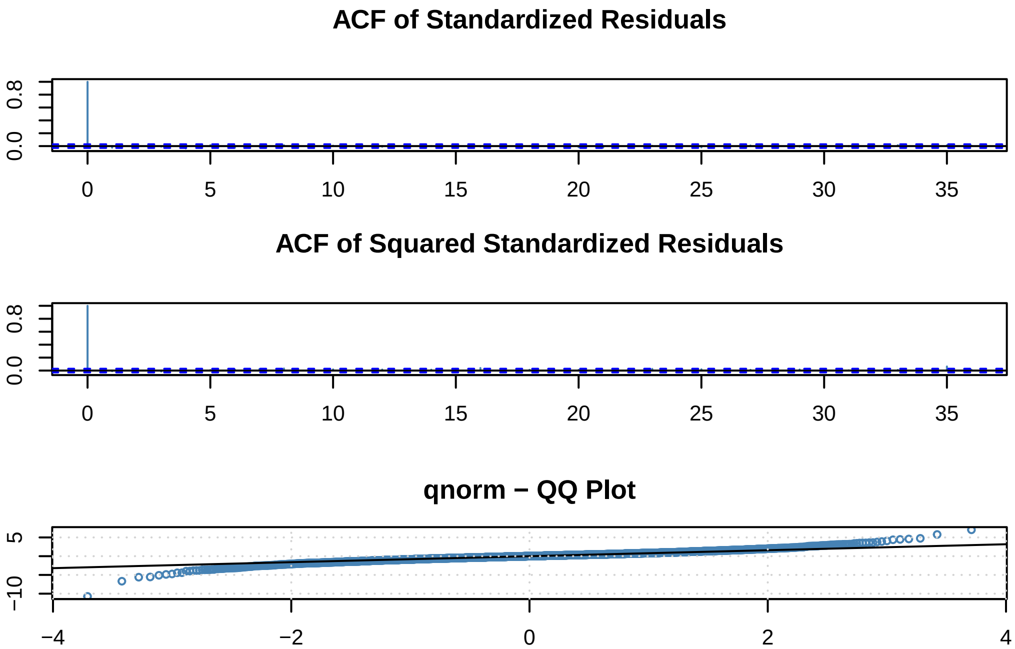
2.4.1 ARCH(1)

```r
# fit a arch(1) [~garch(1,0)] model
arch_1_md <- garchFit(formula = ~ garch(1, 0),
                data = KO$log_returns,
                trace = FALSE)

par(mar=c(3,3,3,1))
tsdiag(arch_1_md)
```



2.4.2 ARCH(3)

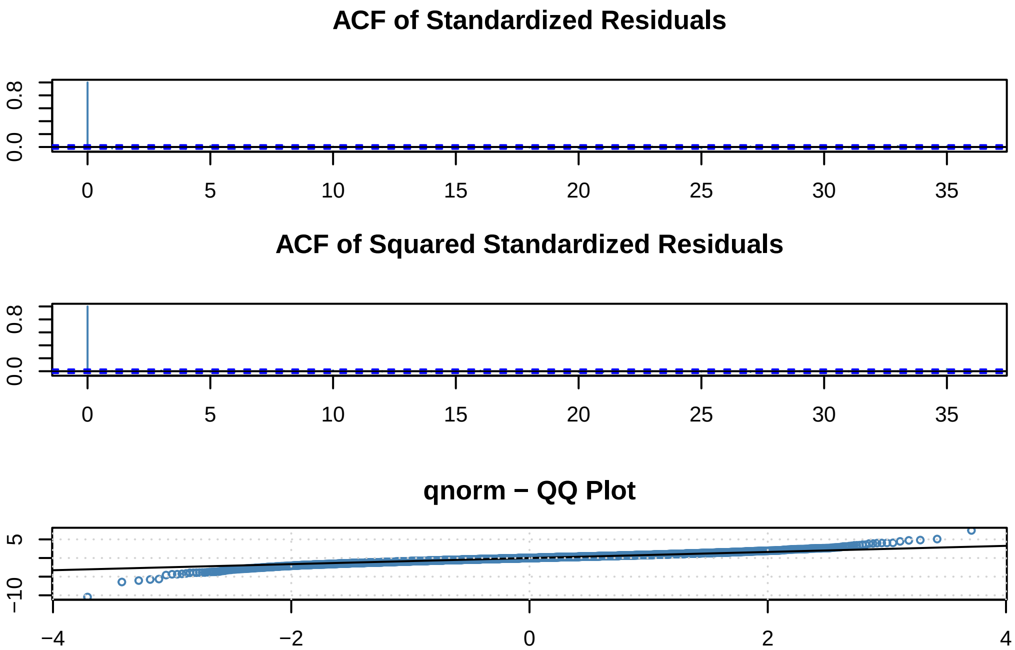
```r
# fit a arch(3) [~garch(3,0)] model
arch_3_md <- garchFit(formula = ~ garch(3, 0),
                      data = KO$log_returns,
                      trace = FALSE)

par(mar=c(3,3,3,1))
tsdiag(arch_3_md)
```



We see that [...]

2.4.3 GARCH(1,1)

```r
# fit a garch(1,1) model
garch_11_md <- garchFit(formula = ~ garch(1, 1),
                        data = KO$log_returns,
                        trace = FALSE)

par(mar=c(3,3,3,1))
tsdiag(garch_11_md)
```



We see that [...]

2.5 Fitted Model Expressions

Task: Write the expression of each fitted model.

Solution

Note that the ARCH(p) model takes the form:

$$\text{Mean Equation } X_t = \mu + Z_t$$

$$\text{Variance Equation } \sigma_t^2 = \alpha_0 + \alpha_1 Z_{t-1}^2 + \alpha_2 Z_{t-2}^2 ... \alpha_p Z_{t-p}^2$$

With the random error Z_t being defined as:

$$Z_t = \sqrt{\sigma_t^2}\varepsilon_t, \quad \varepsilon_t \sim WN(0,1)$$

2.5.1 ARCH(1)

```r
sum1 <- summary(arch_1_md)
```

```
Title:
 GARCH Modelling

Call:
 garchFit(formula = ~garch(1, 0), data = KO$log_returns, trace = FALSE)

Mean and Variance Equation:
 data ~ garch(1, 0)
<environment: 0x000001327a7af508>
 [data = KO$log_returns]

Conditional Distribution:
 norm

Coefficient(s):
        mu       omega      alpha1
6.1419e-04  9.0469e-05  3.4125e-01

Std. Errors:
 based on Hessian

Error Analysis:
        Estimate  Std. Error  t value Pr(>|t|)
mu     6.142e-04   1.444e-04    4.254  2.1e-05 ***
omega  9.047e-05   2.411e-06   37.521  < 2e-16 ***
alpha1 3.413e-01   2.596e-02   13.144  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:
 14958.15    normalized:  3.100777

Description:
 Wed Mar 18 01:39:03 2026 by user: T14


Standardised Residuals Tests:
                                  Statistic    p-Value
 Jarque-Bera Test   R    Chi^2  7946.4809440 0.00000000
 Shapiro-Wilk Test  R    W         0.9440329 0.00000000
 Ljung-Box Test     R    Q(10)    22.8626094 0.01126523
 Ljung-Box Test     R    Q(15)    24.3547456 0.05930900
 Ljung-Box Test     R    Q(20)    30.3436419 0.06447350
 Ljung-Box Test     R^2  Q(10)   230.8443937 0.00000000
 Ljung-Box Test     R^2  Q(15)   337.8879843 0.00000000
 Ljung-Box Test     R^2  Q(20)   408.1074472 0.00000000
 LM Arch Test       R    TR^2    212.3363119 0.00000000

Information Criterion Statistics:
      AIC       BIC       SIC      HQIC
-6.200311 -6.196280 -6.200312 -6.198896
```

```r
# estimates column
sum1$show$matcoef
```

```
           Estimate   Std. Error   t value     Pr(>|t|)
mu     6.141938e-04 1.443931e-04  4.253622 2.103406e-05
omega  9.046889e-05 2.411148e-06 37.521088 0.000000e+00
alpha1 3.412522e-01 2.596197e-02 13.144312 0.000000e+00
```

```r
mu_estimate <- sum1$show$matcoef[1,1]
omega_estimate <- sum1$show$matcoef[2,1]
alpha1_estimate <- sum1$show$matcoef[3,1]
```

Plugging in our estimates, we get a expression of our fitted ARCH(1) model taking the form:

$X_t = 0.000614 + Z_t$

$\sigma_t^2 = 0.000090 + 0.341252 Z_{t-1}^2$

2.5.2 ARCH(3)

```r
sum2 <- summary(arch_3_md)
```

```

Title:
 GARCH Modelling

Call:
 garchFit(formula = ~garch(3, 0), data = KO$log_returns, trace = FALSE)

Mean and Variance Equation:
 data ~ garch(3, 0)
<environment: 0x00000132874b41c8>
 [data = KO$log_returns]

Conditional Distribution:
 norm

Coefficient(s):
        mu       omega      alpha1      alpha2      alpha3
0.00054242  0.00006723  0.12950929  0.18704472  0.14708844

Std. Errors:
 based on Hessian

Error Analysis:
        Estimate  Std. Error  t value Pr(>|t|)
mu     5.424e-04   1.371e-04    3.957 7.59e-05 ***
omega  6.723e-05   2.319e-06   28.991  < 2e-16 ***
alpha1 1.295e-01   1.868e-02    6.933 4.12e-12 ***
alpha2 1.870e-01   2.232e-02    8.382  < 2e-16 ***
alpha3 1.471e-01   1.938e-02    7.590 3.20e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:
 15175.74    normalized:  3.145882
```

```
Description:
 Wed Mar 18 01:39:08 2026 by user: T14


Standardised Residuals Tests:
                                  Statistic    p-Value
 Jarque-Bera Test   R    Chi^2  6773.0145208 0.00000000
 Shapiro-Wilk Test  R    W         0.9586279 0.00000000
 Ljung-Box Test     R    Q(10)    18.3974923 0.04861741
 Ljung-Box Test     R    Q(15)    21.4724589 0.12240104
 Ljung-Box Test     R    Q(20)    26.7136727 0.14351349
 Ljung-Box Test     R^2  Q(10)    14.6745692 0.14438458
 Ljung-Box Test     R^2  Q(15)    25.3977189 0.04485048
 Ljung-Box Test     R^2  Q(20)    36.4103810 0.01375788
 LM Arch Test       R    TR^2     19.6287564 0.07444267

Information Criterion Statistics:
      AIC       BIC       SIC      HQIC
-6.289691 -6.282973 -6.289693 -6.287333
```

```r
sum2$show$matcoef
```

```
           Estimate   Std. Error   t value     Pr(>|t|)
mu     5.424159e-04 1.370779e-04  3.956989 7.590031e-05
omega  6.722972e-05 2.318995e-06 28.990883 0.000000e+00
alpha1 1.295093e-01 1.868052e-02  6.932852 4.124479e-12
alpha2 1.870447e-01 2.231576e-02  8.381732 0.000000e+00
alpha3 1.470884e-01 1.937831e-02  7.590363 3.197442e-14
```

```r
mu_estimate <- sum2$show$matcoef[1,1]
omega_estimate <- sum2$show$matcoef[2,1]
alpha1_estimate <- sum2$show$matcoef[3,1]
alpha2_estimate <- sum2$show$matcoef[4,1]
alpha3_estimate <- sum2$show$matcoef[5,1]
```

Plugging in our estimates, we get a expression of our fitted ARCH(1) model taking the form:

$X_t = 0.00054 + Z_t$

$\sigma_t^2 = 0.00007 + 0.12951 Z_{t-1}^2 + 0.18704 Z_{t-2}^2 + 0.14709 Z_{t-3}^2$

2.5.3 GARCH(1,1)

Note that for GARCH(p,q) models, they take the form:

$$\text{Mean Equation } X_t = \mu + Z_t$$

$$\text{Variance Equation } \sigma_t^2 = \alpha_0 + \alpha_1 Z_{t-1}^2 + \alpha_2 Z_{t-2}^2 + \cdots + \alpha_p Z_{t-p}^2 + \beta_1 \sigma_{t-1}^2 + \beta_2 \sigma_{t-2}^2 + \cdots + \beta_q \sigma_{t-q}^2$$

```r
sum3<- summary(garch_11_md)
```

```
Title:
 GARCH Modelling

Call:
```

```
 garchFit(formula = ~garch(1, 1), data = KO$log_returns, trace = FALSE)

Mean and Variance Equation:
 data ~ garch(1, 1)
<environment: 0x0000013284974d58>
 [data = KO$log_returns]

Conditional Distribution:
 norm

Coefficient(s):
        mu       omega      alpha1       beta1
4.9355e-04  3.1741e-06  6.6709e-02  9.0752e-01

Std. Errors:
 based on Hessian

Error Analysis:
        Estimate  Std. Error  t value Pr(>|t|)
mu     4.935e-04   1.355e-04    3.642 0.000271 ***
omega  3.174e-06   6.618e-07    4.796 1.62e-06 ***
alpha1 6.671e-02   7.844e-03    8.505  < 2e-16 ***
beta1  9.075e-01   1.189e-02   76.337  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log Likelihood:
 15265.26    normalized:  3.16444

Description:
 Wed Mar 18 01:39:13 2026 by user: T14



Standardised Residuals Tests:
                                   Statistic    p-Value
 Jarque-Bera Test   R    Chi^2  6400.4147514 0.00000000
 Shapiro-Wilk Test  R    W         0.9603192 0.00000000
 Ljung-Box Test     R    Q(10)    17.9134462 0.05644135
 Ljung-Box Test     R    Q(15)    19.5904585 0.18821227
 Ljung-Box Test     R    Q(20)    23.8661546 0.24828773
 Ljung-Box Test     R^2  Q(10)     3.9874863 0.94790973
 Ljung-Box Test     R^2  Q(15)     4.5235915 0.99544926
 Ljung-Box Test     R^2  Q(20)     5.9770498 0.99892815
 LM Arch Test       R    TR^2      4.1182000 0.98120774

Information Criterion Statistics:
      AIC       BIC       SIC      HQIC
-6.327221 -6.321847 -6.327222 -6.325334
```

```
sum3$show$matcoef
```

```
           Estimate   Std. Error   t value     Pr(>|t|)
mu     4.935481e-04 1.355299e-04  3.641619 2.709292e-04
omega  3.174092e-06 6.617873e-07  4.796243 1.616692e-06
```

```
alpha1 6.670870e-02 7.843569e-03  8.504891 0.000000e+00
beta1  9.075201e-01 1.188836e-02 76.336888 0.000000e+00
```

```r
mu_estimate <- sum3$show$matcoef[1,1]
omega_estimate <- sum3$show$matcoef[2,1]
alpha1_estimate <- sum3$show$matcoef[3,1]
beta1_estimate <- sum3$show$matcoef[4,1]
```

Plugging in our estimates, we get a expression of our fitted ARCH(1) model taking the form:

$X_t = 0.00049 + Z_t$

$\sigma_t^2 = 0.00000 + 0.06671 Z_{t-1}^2 + 0.90752 \sigma_{t-1}^2$

2.6 Results & Best Fit

Task: Analyze the diagnostic results and determine which of the three models provides the best fit

Solution

To decide which of the three models gives the best fit, we will compare information criteria (AIC/BIC), significance of coefficients, and robustness checks using Ljung-Box, Shapiro-Wilk, and Jarque-Bera tests.

2.6.1 Information Criteria

```r
Models = c('ARCH(1)', 'ARCH(3)', 'GARCH(1,1)')

df_results <- rbind(sum1$ics,
                    sum2$ics,
                    sum3$ics)

df_results <- as.data.frame(df_results)

rownames(df_results) <- Models
df_results # table of results
```

```
                 AIC       BIC       SIC      HQIC
ARCH(1)    -6.200311 -6.196280 -6.200312 -6.198896
ARCH(3)    -6.289691 -6.282973 -6.289693 -6.287333
GARCH(1,1) -6.327221 -6.321847 -6.327222 -6.325334
```

It is important we only compare models using the same IC for each other; thus, we will consider all three model's AIC scores, and then all of their BIC scores:

```r
# aic
df_results %>%
  select(AIC) %>%
  arrange(AIC)
```

```
                 AIC
GARCH(1,1) -6.327221
ARCH(3)    -6.289691
ARCH(1)    -6.200311
```

```r
# bic
df_results %>%
  select(BIC) %>%
  arrange(BIC)
```

```
                 BIC
GARCH(1,1) -6.321847
ARCH(3)    -6.282973
ARCH(1)    -6.196280
```

For both information criteria, the GARCH(1,1) has the lowest score, which is preferred as it indicates the best balance of fit and parsimony.

2.6.2 Coefficient Significance

```
# ARCH(1)
sum1$show$matcoef
```

```
           Estimate   Std. Error   t value     Pr(>|t|)
mu     6.141938e-04 1.443931e-04  4.253622 2.103406e-05
omega  9.046889e-05 2.411148e-06 37.521088 0.000000e+00
alpha1 3.412522e-01 2.596197e-02 13.144312 0.000000e+00
```

```
# ARCH(3)
sum2$show$matcoef
```

```
           Estimate   Std. Error   t value     Pr(>|t|)
mu     5.424159e-04 1.370779e-04  3.956989 7.590031e-05
omega  6.722972e-05 2.318995e-06 28.990883 0.000000e+00
alpha1 1.295093e-01 1.868052e-02  6.932852 4.124479e-12
alpha2 1.870447e-01 2.231576e-02  8.381732 0.000000e+00
alpha3 1.470884e-01 1.937831e-02  7.590363 3.197442e-14
```

```
# GARCH(1,1)
sum3$show$matcoef
```

```
           Estimate   Std. Error   t value     Pr(>|t|)
mu     4.935481e-04 1.355299e-04  3.641619 2.709292e-04
omega  3.174092e-06 6.617873e-07  4.796243 1.616692e-06
alpha1 6.670870e-02 7.843569e-03  8.504891 0.000000e+00
beta1  9.075201e-01 1.188836e-02 76.336888 0.000000e+00
```

For all three models, we do not contain any non-significant coefficients, suggesting that all three models fit the data well and that including more time lags is successful in helping to explain more of the data.

2.6.3 Ljung-Box Test

Furthermore, analyzing the Ljung-Box test on the squared standardized residuals for Model 3 yielded a p-value of 0.XX, confirming that the model successfully captured all conditional heteroskedasticity (ARCH effects). While Model 2 also passed the residual diagnostics, it had higher information criteria and a non-significant parameter, making Model 3 the superior fit.

```
# ARCH(1)
sum1$stat_tests[3:5,]
```

```
                                Statistic    p-Value
 Ljung-Box Test     R    Q(10)   22.86261 0.01126523
 Ljung-Box Test     R    Q(15)   24.35475 0.05930900
 Ljung-Box Test     R    Q(20)   30.34364 0.06447350
```

```
sum1$stat_tests[6:8,]
```

```
                                Statistic p-Value
 Ljung-Box Test     R^2  Q(10)   230.8444       0
 Ljung-Box Test     R^2  Q(15)   337.8880       0
```

```
 Ljung-Box Test     R^2  Q(20)   408.1074       0
```

```
# ARCH(3)
sum2$stat_tests[3:5,]
```

```
                                Statistic    p-Value
 Ljung-Box Test     R    Q(10)   18.39749 0.04861741
 Ljung-Box Test     R    Q(15)   21.47246 0.12240104
 Ljung-Box Test     R    Q(20)   26.71367 0.14351349
```

```
sum2$stat_tests[6:8,]
```

```
                                Statistic    p-Value
 Ljung-Box Test     R^2  Q(10)   14.67457 0.14438458
 Ljung-Box Test     R^2  Q(15)   25.39772 0.04485048
 Ljung-Box Test     R^2  Q(20)   36.41038 0.01375788
```

```
# GARCH(1,1)
sum3$stat_tests[3:5,]
```

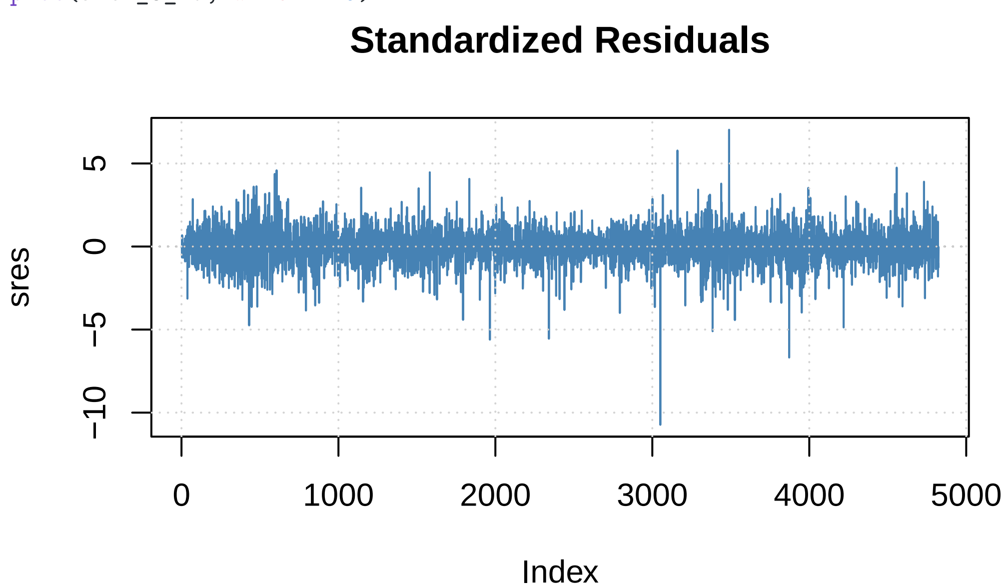
```
                                Statistic    p-Value
 Ljung-Box Test     R    Q(10)   17.91345 0.05644135
 Ljung-Box Test     R    Q(15)   19.59046 0.18821227
 Ljung-Box Test     R    Q(20)   23.86615 0.24828773
```

```
sum3$stat_tests[6:8,]
```

```
                                Statistic   p-Value
 Ljung-Box Test     R^2  Q(10)   3.987486 0.9479097
 Ljung-Box Test     R^2  Q(15)   4.523592 0.9954493
 Ljung-Box Test     R^2  Q(20)   5.977050 0.9989281
```

We see that [...]

```
# Interactive plotting menu
plot(arch_3_md, which = 9)
```



We see that [...]

2.6.4 Jarque-Beta/Shapiro-Wilk

```r
# ARCH(1)
sum1$stat_tests[1:2,]
```

```
                                  Statistic p-Value
 Jarque-Bera Test   R    Chi^2  7946.4809440       0
 Shapiro-Wilk Test  R    W         0.9440329       0
```

```r
# ARCH(3)
sum2$stat_tests[1:2,]
```

```
                                  Statistic p-Value
 Jarque-Bera Test   R    Chi^2  6773.0145208       0
 Shapiro-Wilk Test  R    W         0.9586279       0
```

```r
# GARCH(1,1)
sum3$stat_tests[1:2,]
```

```
                                  Statistic p-Value
 Jarque-Bera Test   R    Chi^2  6400.4147514       0
 Shapiro-Wilk Test  R    W         0.9603192       0
```

2.6.5 LM Arch Test

The LM Arch test is a hypothesis test under the null that H_0 : autocorrelated squared residuals. If we reject the null at the 5% significance level ($\alpha = 0.05$), then then this suggests that autoregressive conditional heteroskedasticity exists in our model, and we should find a better model to capture more of the volatility structure. If we fail to reject the null, then

```r
# ARCH(1)
sum1$stat_tests[9,]
```

```
Statistic   p-Value
 212.3363    0.0000
```

```r
# ARCH(3)
sum2$stat_tests[9,]
```

```
  Statistic     p-Value
19.62875637  0.07444267
```

```r
# GARCH(1,1)
sum3$stat_tests[9,]
```

```
Statistic   p-Value
4.1182000 0.9812077
```

2.6.6 Model Decision